

Lab 3: Component Modelling & Architectural Pattern Selection

Written Justification — Architectural Decision
System: Digital Gift Card & Loyalty Point Exchange | Problem Statement #34

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Architecture Selection

We chose Microservices Architecture for the Digital Gift Card & Loyalty Point Exchange System.

1. Architectural Choice: A Microservices Architecture was selected over Layered and Client-Server alternatives. The platform is decomposed into independently deployable services — Account Holder Portal, Loyalty Aggregation Service, Point Conversion Engine, Voucher & Gift Card Service, Merchant Integration Service, and a shared Ledger Database — each owning a single, well-defined responsibility and communicating through explicit provided/required interfaces (see Component Diagram).

2. Two Scenario-Related Reasons:

a) Independent scaling under uneven load: Loyalty aggregation, point conversion, and voucher redemption see very different and unpredictable traffic — e.g., redemption spikes during promotional events — so each service can be scaled horizontally on its own instead of over-provisioning the entire monolith, directly supporting the 250 ms processing target under peak load (NFR-001).

b) Independent deployment and fault isolation: Merchant Partner integrations change frequently (new partners, rate updates) and anti-fraud voucher logic needs rapid patching. Microservices let each team ship and roll back these services independently, and a failure or slowdown in one service (e.g., Merchant Integration) does not cascade into or take down redemption or payment-adjacent flows.

3. Security Advantage: Decomposing the system lets the Voucher & Gift Card Service — the component that performs single-use voucher generation and anti-fraud locking — sit behind its own hardened service boundary with its own authentication, authorization, and audit logging. This narrows the blast radius of any single breach: an attacker who compromises the Merchant Integration Service, for example, cannot directly reach the Ledger Database or forge a voucher, since every cross-service call passes through an explicit, independently-secured interface rather than shared in-process memory.

4. Performance Benefit: Because point conversion and gift-card redemption must complete within 250 ms with zero ledger-balance drift (NFR-001), the Point Conversion Engine and Voucher & Gift Card Service can be deployed on dedicated, right-sized, horizontally-scaled infrastructure, independent of lighter-weight services such as Merchant Integration. This targeted resource allocation keeps the latency-critical path fast even during peak load, which a single-tiered layered or client-server design could not achieve without scaling the whole application.

Reference: Component Diagram (component_diagram.png) submitted alongside this document shows all 6 components and their provided/required interfaces.